Meningococcemia (MONDO:0042972): A Comprehensive Disease Characteristics Report

Category: Infectious Disease | Causative agent: Neisseria meningitidis (NCBI:txid487) Evidence base: Aggregated disease-level literature (reviews, cohort studies, surveillance, mechanistic in vitro / humanized-mouse studies). No patient-level EHR data were used.

Summary

Meningococcemia is an acute, frequently fulminant bloodstream infection caused by the human-restricted Gram-negative diplococcus *Neisseria meningitidis*. It is one of the two principal clinical forms of invasive meningococcal disease (IMD), the other being meningococcal meningitis; the two frequently coexist. Meningococcemia is distinguished by rapid progression from a nonspecific febrile prodrome to septic shock, disseminated intravascular coagulation (DIC), purpura fulminans, bilateral adrenal hemorrhage (Waterhouse-Friderichsen syndrome), and multi-organ failure, often within 24 hours of first symptoms.

The pathophysiology is now understood at molecular resolution. Circulating meningococci adhere to microvascular endothelium via **type IV pili**, triggering **ADAM10-dependent shedding of the Endothelial Protein C Receptor (EPCR)**, which cripples the anticoagulant Protein C pathway and produces the localized microvascular thrombosis characteristic of purpura fulminans. In parallel, the bacterium's **lipooligosaccharide/lipopolysaccharide (LPS/endotoxin)** drives a TLR4-mediated cytokine storm (IL-10, TNF, MIF) whose magnitude scales with bacterial load and predicts mortality. Host susceptibility is dominated by **complement function**: inherited deficiencies of the terminal pathway (C5–C9), properdin (CFP, X-linked), and factor D (CFD) markedly increase risk, and this phenotype is now iatrogenically reproduced by the anti-C5 monoclonal antibodies eculizumab and ravulizumab.

Despite prompt third-generation cephalosporin therapy, meningococcemia carries a case-fatality rate of roughly **8–17%**, rising above 20% in the fulminant hemodynamic form, and leaves approximately **10–20% of survivors** with permanent sequelae (limb amputation, hearing loss, skin scarring, renal dysfunction, neurologic deficits). The disease is preventable: conjugate vaccines (MenAfriVac, quadrivalent ACWY, pentavalent ACWYX) and the protein-based 4CMenB vaccine have transformed epidemiology, and rapid contact chemoprophylaxis with ciprofloxacin or rifampicin interrupts secondary transmission. This report synthesizes 14 confirmed findings drawn from 57 reviewed papers across all 15 requested disease-characteristic domains.

Section 1 — Disease Information

Overview. Meningococcemia (meningococcal septicemia) is the bloodstream-infection presentation of invasive meningococcal disease caused by *Neisseria meningitidis*, an encapsulated Gram-negative diplococcus that is an obligate human commensal of the nasopharynx. Invasion from the mucosal reservoir into the blood is rare relative to carriage, but when it occurs it can produce fulminant sepsis. IMD classically presents in two overlapping clinical forms — a **neurological/meningitic form** and a **hemodynamic/septicemic form (meningococcemia)** — with a nationwide French adult ICU cohort (RETRO-MENINGO, 654 patients) finding 62% neurological and 38% hemodynamic presentations (PMID: 40794167).

Key identifiers.

Resource	Identifier
Mondo	MONDO:0042972 (meningococcemia)
MeSH	Meningococcal Infections; *Neisseria meningitidis*
ICD-10	A39.2 (acute meningococcemia); A39.4 (meningococcemia, unspecified); A39.1 (Waterhouse-Friderichsen syndrome)
ICD-11	1D01 (meningococcal infection / invasive meningococcal disease)
NCBI Taxonomy	*Neisseria meningitidis* txid487

Synonyms/alternative names. Meningococcal septicemia; meningococcal sepsis; meningococcal bacteremia; invasive meningococcal disease (septicemic form); systemic meningococcal disease. *Purpura fulminans* and *Waterhouse-Friderichsen syndrome* denote severe complications rather than synonyms.

Nature of the evidence. Meningococcemia is characterized primarily through **aggregated disease-level resources** — national surveillance systems, prospective cohorts, autopsy series, and reference-laboratory serogroup/genotype data — supplemented by individual case reports for rare complications. It is not a Mendelian/EHR-genotype disease; host-genetic contribution is confined to susceptibility modifiers (below).

Section 2 — Etiology

Primary cause (infectious). The sole causal agent is *Neisseria meningitidis*. Twelve serogroups are defined by capsular polysaccharide; six (A, B, C, W, X, Y) cause nearly all invasive disease. Disease results from the interaction of exposure to a virulent strain with absence of host resistance (PMID: 11057965).

Genetic host risk factors. Inherited deficiencies of the **terminal complement components (C5, C6, C7, C8, C9)**, **properdin (CFP; X-linked)**, and **factor D (CFD)** of the alternative pathway strongly predispose to invasive meningococcal disease, often with recurrent episodes and atypical serogroups (Y, W-135) (F001). *"Individuals, deficient of properdin or of the terminal pathway components (C5 to C9), are highly susceptible to meningococcal disease"* (PMID: 25037634). Factor D deficiency *"abolishes alternative-pathway dependent complement activation by N meningitidis, and leads to an increased susceptibility to invasive meningococcal disease"* (PMID: 16527897). Combined **properdin and mannose-binding lectin (MBL2)** deficiency shows epistatic elevation of risk (PMID: 16337490).

Environmental risk factors. Nasopharyngeal **carriage** is the reservoir; risk of acquisition and disease is elevated by **active and passive smoking** (personal OR 2.5, passive OR 1.6), **male sex** (OR 1.7), **age 15–24 years**, **household crowding**, and **close/household contact** with a case (PMID: 11399012) (F007). Infancy and immunosuppression are additional risk states; the hemodynamic form is associated with older age and immunosuppression (PMID: 40794167).

Protective factors. Carriage of the commensal *Neisseria lactamica* in early childhood is associated with natural immunity/protection — disease *"is a reflection of ... an absence of host resistance among those individuals not carrying N. lactamica"* (PMID: 11057965). Vaccination (Section 13) is the dominant modifiable protective factor.

Gene–environment interaction. Susceptibility is fundamentally a G×E product: a complement-deficient (or complement-inhibited) host who acquires a virulent, non-carried strain from a crowded/smoking environment is at greatly amplified risk. The iatrogenic version — anti-C5 therapy plus meningococcal exposure — is the clearest example (Section 9/F009).

Section 3 — Phenotypes

Meningococcemia's phenotype evolves rapidly. The classic early triad of **leg pain, cold hands/feet, and abnormal skin color** often precedes the hemorrhagic rash and meningeal signs. In primary-care children, five features carry clinically useful positive likelihood ratios (F012, PMID: 21375891):

Phenotype	HPO suggestion	LR+ (95% CI)	Type
Confusion	HP:0001289	24.2 (11.5–51.3)	Sign/behavioral
Leg pain	HP:0031057 (limb pain)	7.6 (4.9–11.9)	Symptom
Photophobia	HP:0000613	6.5 (3.8–11.0)	Symptom
Rash (petechial/purpuric)	HP:0000979 / HP:0000967	5.5 (4.3–7.1)	Physical manifestation
Neck pain/stiffness	HP:0031179 / HP:0031352	5.3 (3.5–8.3)	Sign

Additional phenotypes and suggested HPO terms:

- **Fever** (HP:0001945) — near-universal nonspecific prodrome.
- **Non-blanching petechial/purpuric rash → purpura fulminans** (HP:0011000 purpura fulminans; HP:0000979 purpura) — may begin maculopapular, becomes hemorrhagic/non-blanching; the hallmark of meningococcemia.
- **Septic shock / hypotension** (HP:0031273 shock; HP:0002615 hypotension) — the dominant driver of mortality (aOR 23.3, PMID: 25605282).

- **Disseminated intravascular coagulation** (HP:0005521) — consumptive coagulopathy.
- **Cold extremities / poor perfusion** (HP:0025247 cold extremities) — early red flag.
- **Distal ischemia/necrosis, gangrene** (HP:0100758 gangrene) — leads to amputation.
- **Adrenal hemorrhage / acute adrenal insufficiency** (HP:0008207) — Waterhouse-Friderichsen syndrome.
- **Myocarditis / cardiac dysfunction** (HP:0001638-adjacent) — reversible myocardial depression documented in case reports (PMID: 30871501, PMID: 30293563).
- **Laboratory abnormalities:** elevated LPS/endotoxin and IL-10, thrombocytopenia, coagulopathy, elevated CSF lactate even without pleocytosis (PMID: 30871501).

Onset/severity/progression. Onset is **acute**, spanning all ages but concentrated in infants and adolescents/young adults; severity ranges **moderate-to-severe/fulminant**; progression is **rapid and episodic** (single acute episode, though complement-deficient individuals suffer recurrences).

Quality-of-life impact. Survivors of the septicemic form bear substantial long-term QoL burden — amputation, scarring, hearing loss, renal dysfunction, and persistent emotional distress in patients and caregivers (PMID: 40627264).

Section 4 — Genetic/Molecular Information

This is not a Mendelian disease of a single causal human gene. The relevant human genetics concern **host susceptibility modifiers**, all in the complement system:

Gene	Locus / pathway	Consequence	Effect
C5, C6, C7, C8A/B, C9	terminal pathway	Loss of function → no membrane attack complex	High susceptibility, recurrent IMD, atypical serogroups (PMID: 25037634)
CFP (properdin)	Xp11.23; X-linked	Loss of alternative-pathway amplification	Severe/recurrent IMD; serogroups Y, W-135 (PMID: 16337490, PMID: 10909851)
CFD (factor D)	alternative pathway	Abolishes alternative-pathway activation vs *N. meningitidis*	Increased IMD susceptibility (PMID: 16527897)
MBL2	lectin pathway	Reduced MBL	Modifier; epistatic with properdin (PMID: 16337490)

Variant characteristics. Properdin type I deficiency arises from diverse loss-of-function variants — missense substitutions concentrated in conserved residues of exons 7–8 and premature stops elsewhere in the properdin gene (PMID: 10909851); factor D deficiency was described from two novel *CFD* mutations segregating with meningococcal infection in a family (PMID: 16527897). These are **germline, loss-of-function** variants; classification is pathogenic for the deficiency phenotype. Population allele frequencies are low; properdin deficiency is X-linked (males affected, carrier females show skewed X-inactivation, PMID: 10909851).

Pathogen genetics (the true "causal" genome). Virulence determinants of *N. meningitidis* include the polysaccharide **capsule** (serogroup-defining, e.g. *siaD*/*ctrA* loci), **type IV pili** (PilE, and the ATPases PilF/PilT governing pilus assembly/retraction), and **lipooligosaccharide (LOS/LPS)**. Antimicrobial-resistance variants include **penA** mutations (reduced penicillin susceptibility) and **gyrA** mutations (ciprofloxacin resistance) (PMID: 26357666, PMID: 35467371).

Epigenetics / chromosomal abnormalities. Not applicable as disease drivers in the human host.

Section 5 — Environmental Information

Infectious agent. *Neisseria meningitidis* (NCBI Taxonomy txid487), a strictly human-restricted, encapsulated Gram-negative diplococcus, transmitted person-to-person via respiratory droplets and secretions. Serogroups A, B, C, W, X, Y dominate invasive disease; serogroup distribution varies geographically and shifts over time (serogroup replacement; Section 9).

Lifestyle factors. Smoking (active and passive) is the best-established modifiable environmental risk factor, elevating carriage/acquisition (personal OR 2.5, passive OR 1.6) (PMID: 11399012). Bar/party attendance and shared bedrooms also associate with carriage (PMID: 27861618).

Physical/social environment. Crowding (people per bedroom), **household close contact**, and **congregate settings** (dormitories, schools, military barracks) increase transmission (PMID: 27861618, PMID: 11057965). No chemical toxin, radiation, or occupational exposure is causal; CTD/EPA-type environmental toxins are not applicable.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain (initiating event → clinical manifestation)

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1. Nasopharyngeal acquisition of a virulent, encapsulated N. meningitidis strain
     → colonizes mucosa (carriage).
2. In a susceptible host (absent bactericidal antibody / deficient or inhibited
   complement), meningococci breach the mucosa and enter the bloodstream
     → RESULTS IN bacteremia (meningococcemia).
3. Deficient terminal/alternative complement (inherited C5–C9, properdin, factor D,
   OR iatrogenic anti-C5) fails to form the membrane attack complex
     → RESULTS IN failure to lyse the meningococcus → sustained/high-grade bacteremia.
4. Type IV pili mediate adhesion of circulating meningococci to human microvascular
   endothelial cells
     → LEADS TO vascular colonization; pilus retraction (PilT) sustains bacteremia.
        [BRANCH A — coagulopathy]                    [BRANCH B — inflammation]
5A. Adhesion triggers ADAM10-dependent          5B. LOS/LPS (lipid A) engages TLR4 on
    shedding of Endothelial Protein C Receptor       monocytes/endothelium
    (EPCR)                                             → RESULTS IN NF-κB activation and a
     → RESULTS IN loss of Protein C activation.          cytokine storm (TNF, IL-10, MIF,
6A. Impaired activated Protein C anticoagulant       IL-1β), scaling with bacterial/LPS load.
    activity + endothelial injury                6B. Cytokines + complement anaphylatoxins
     → LEADS TO localized microvascular thrombosis     → RESULT IN vasodilation, capillary leak
        and fibrin deposition (purpura fulminans).      and a circulating myocardial depressant
                                                          factor → distributive + cardiogenic sh
        ↓  (branches converge)
7. Disseminated intravascular coagulation (consumption of clotting factors/platelets)
   + microthrombosis + capillary leak
     → RESULTS IN tissue hypoperfusion, purpura, distal ischemia/gangrene, and
        bilateral adrenal hemorrhage (Waterhouse-Friderichsen syndrome).
8. Multi-organ hypoperfusion + endotoxin/cytokine injury
     → RESULTS IN septic shock, acute kidney injury, myocardial dysfunction,
        and — if untreated — death.
9. In survivors, ischemic/thrombotic tissue loss
     → LEADS TO amputation, skin scarring, hearing loss, and neurologic sequelae.
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(Steps 5A–6A demonstrated in vitro and in humanized mouse models; step 5B/6B the TLR4-cytokine arm is supported by human plasma correlations. The convergence on DIC/shock and organ damage is established clinically.)

Detail by category

Molecular pathways. (i) **Complement cascade** — alternative and terminal pathways; membrane attack complex (C5b-9) is the effector normally clearing the meningococcus; its loss (genetic or pharmacologic) is the master susceptibility lesion (F001, F009). (ii) **TLR4/MyD88–NF-κB signaling** downstream of LPS/lipid A drives cytokine transcription (F005). (iii) **Protein C anticoagulant pathway** — EPCR/thrombomodulin-dependent; its collapse via EPCR shedding is the specific coagulopathic lesion of meningococcal purpura fulminans (F002). Suggested GO terms: GO:0006956 (complement activation), GO:0006957 (alternative pathway), GO:0002250 (adaptive immune response), GO:0007249 (I-κB kinase/NF-κB signaling), GO:0007596 (blood coagulation), GO:0030194 (positive regulation of blood coagulation).

Cellular processes. Inflammation (GO:0006954), endothelial activation and injury, leukocyte recruitment, platelet activation, and endothelial EPCR shedding. Cell types (CL): vascular endothelial cell (CL:0000115), monocyte (CL:0000576)/macrophage (CL:0000235), neutrophil (CL:0000775), platelet (CL:0000233).

Protein dysfunction. ADAM10 metalloprotease cleaves EPCR from the endothelial surface (*"we identified ADAM10 (A Disintegrin And Metalloproteinase-10) as the protease responsible for this shedding"*, PMID: 29630665); this is a gain of proteolytic activity producing loss of anticoagulant receptor function.

Immune involvement. Two-edged: complement deficiency/inhibition is immunodeficiency that permits invasion, while overwhelming LPS-driven cytokine release is a maladaptive hyper-inflammatory response. IL-10 is a dominant, paradoxically deleterious mediator — IL-10 immunodepletion from meningococcal sepsis plasma markedly changes monocyte gene expression and cytokine release (PMID: 25233959); MIF is highest in shock (PMID: 17438452).

Tissue damage mechanisms. Ischemia/thrombosis (microvascular occlusion), hemorrhage (adrenal, skin), and necrosis (distal digits/limbs). A circulating **10–25 kDa proteinaceous myocardial depressant factor** causes reversible cardiac dysfunction (PMID: 12394943).

Molecular profiling (severity biomarkers). Plasma LPS and IL-10 correlate with severity and are highest in non-survivors (PMID: 25233959); in shock vs non-shock patients, median LPS 105 vs <0.04 EU/mL, IL-10 2.1×10^{4} vs 4×10^{1} pg/mL, and meningococcal DNA 2.9×10^{7} vs $<10^{3}$ copies/mL (PMID: 18679115) (F005).

Section 7 — Anatomical Structures Affected

Organ / system level. - **Primary:** blood/circulatory system (UBERON:0000178 blood; UBERON:0001981 blood vessel), **microvascular endothelium** (skin dermal microvessels are the demonstrated adhesion target). - **Skin** (UBERON:0002097) — purpura fulminans, petechiae, necrosis. - **Adrenal glands** (UBERON:0002369) — bilateral hemorrhage (Waterhouse-Friderichsen syndrome). - **Heart** (UBERON:0000948) — myocarditis/myocardial depression. - **Kidneys** (UBERON:0002113) — acute kidney injury/renal dysfunction. - **Meninges/CNS** (UBERON:0002360) — when concurrent meningitis is present. - **Secondary:** distal extremities (ischemia/amputation), lungs (ARDS), liver, and coagulation system (DIC). Body systems: cardiovascular, hematologic/immune, integumentary, endocrine, renal, and nervous.

Tissue and cell level. Endothelial tissue is the specific target; **vascular endothelial cells** (CL:0000115) are directly colonized by type IV pili. Immune effectors: monocytes/macrophages (CL:0000576/CL:0000235), neutrophils (CL:0000775), platelets (CL:0000233).

Subcellular level. Endothelial **cell surface/plasma membrane** (GO:0005886) — site of EPCR and its ADAM10-mediated shedding; **extracellular space** (GO:0005615) — cytokines, LPS, complement, shed EPCR.

Localization / lateralization. Purpura is typically **bilateral and widespread** (trunk, extremities); adrenal hemorrhage is characteristically **bilateral**; distal ischemia often symmetric ("symmetric peripheral gangrene").

Section 8 — Temporal Development

Onset. Acute to hyperacute. Age distribution is **bimodal-to-trimodal** with peaks in infants (<1 yr), adolescents/young adults (15–24 yr), and older adults; onset can occur at any age.

Progression. From nonspecific febrile prodrome the disease can progress to shock and death within ~24 hours. Stages: (1) prodrome (fever, malaise, myalgia); (2) early sepsis red-flags (leg pain, cold hands/feet, mottled skin) (PMID: 21375891); (3) fulminant phase (purpura fulminans, DIC, shock, adrenal hemorrhage); (4) recovery or death. Course is **acute and self-limited to a single episode** in immunocompetent hosts, but **recurrent** in complement-deficient/inhibited patients.

Patterns. Remission is treatment-induced (antibiotics + supportive care) in survivors; there is no chronic phase. The **critical window for intervention is extremely short** — early recognition and immediate antibiotics before shock is the principal determinant of survival, since shock is the dominant independent predictor of death (aOR 23.3) (PMID: 25605282).

Section 9 — Inheritance and Population

Epidemiology. Endemic incidence in temperate high-income regions is typically **<1–2 per 100,000 per year**, far higher during African meningitis-belt epidemics. US population mortality (1990–2002, 3,335 deaths) was age-adjusted **0.10/100,000/yr**, with 58% of deaths in persons <25 years and elevated mortality in infants, young adults (15–24), and the elderly (>74) (PMID: 16511378) (F014). Outbreak incidence can spike (e.g., middle-school students 6.57/100,000, PMID: 18322129).

Age distribution. Infants <1 yr are the largest single pediatric group (38.6%); children <5 account for 62.7% of pediatric cases (PMID: 42802197). Adolescents/young adults are the key **carriage reservoir** (PMID: 42702664).

Seasonality. Winter/early-spring peaks; mortality *"rose in winter months and declined during the summer"* (PMID: 16511378).

Sex ratio. Male predominance in carriage (OR 1.7) (PMID: 11399012).

Demographic disparities. African-American mortality was **1.45**× white and **3.32**× Asian/Pacific Islander rates (PMID: 16511378).

Geographic distribution / serogroup dynamics. The **African meningitis belt** historically suffered serogroup A epidemics, now eliminated by MenAfriVac; non-A serogroups (C, W, X) have emerged (serogroup X rose significantly in Ghana, P<0.01) (PMID: 37024833). Serogroup B predominates in Europe/Americas; W (ST-11) and Y circulate widely.

Inheritance of susceptibility. The disease itself is infectious, not inherited, but host susceptibility follows the inheritance of complement genes: **X-linked** for properdin (CFP), **autosomal recessive** for terminal-component and factor D deficiencies. Carrier females of properdin deficiency show variable properdin levels correlated with X-inactivation skewing (PMID: 10909851).

Section 10 — Diagnostics

Laboratory / microbiology. Definitive diagnosis rests on **culture of *N. meningitidis* from blood or CSF** (sterile-site isolation) plus **real-time PCR** (F011). Multiplex 5′-nuclease (TaqMan) PCR targeting the capsular transport gene **ctrA** detects the organism in CSF, blood, plasma, and serum with ~**88% sensitivity** versus culture (*"The respective sensitivities (as percentages) for N. meningitidis, H. influenzae, and S. pneumoniae were 88.4, 100, and 91.8"*, PMID: 11283086); optimized extraction raised ctrA/siaD sensitivity to 93%/91%, and PCR introduction produced a **56% increase in laboratory-confirmed cases** in the UK (PMID: 10799809). **siaD PCR** distinguishes serogroups B and C. Supportive labs: thrombocytopenia, prolonged PT/PTT, low fibrinogen, elevated D-dimer (DIC panel), lactate, procalcitonin/CRP, and low cortisol in WFS.

Biomarkers of severity. Plasma LPS, IL-10, MIF, and meningococcal DNA copy number stratify shock risk and mortality (PMID: 25233959, PMID: 18679115, PMID: 17438452).

A diagnostic pearl: in fulminant septicemia meningococci may be present in CSF without pleocytosis or biochemical inflammation; **elevated CSF lactate** can be the key clue and should prompt meningeal-dose beta-lactam (PMID: 30871501).

Imaging. Adrenal imaging (CT/US) for suspected Waterhouse-Friderichsen syndrome; echocardiography for myocardial dysfunction; imaging is adjunctive, not diagnostic of the infection.

Clinical criteria / differential diagnosis. Diagnosis is clinical–microbiological. Key differentials: other bacterial sepsis with purpura (pneumococcal, *H. influenzae*, Gram-negative DIC), Rocky Mountain spotted fever and other rickettsioses, viral hemorrhagic fevers, thrombotic microangiopathies (aHUS/TTP), immune thrombocytopenia, and non-infectious purpura fulminans (protein C/S deficiency). The non-blanching purpuric rash plus rapid shock in a febrile child/young adult is the classic clinical gestalt (PMID: 21375891).

Genetic testing. Not for diagnosis of the acute infection, but **complement functional testing (CH50/AH50) and targeted sequencing of terminal-complement, CFP, CFD, MBL2 genes** are indicated after invasive disease — especially recurrent episodes, atypical serogroups (Y/W-135), or family history — to identify predisposing deficiency (PMID: 25037634, PMID: 31421540).

Section 11 — Outcome / Prognosis

Mortality. Case-fatality ranges roughly **8–17%** overall, higher in the fulminant septicemic form. Canadian prospective cohort (n=868, 2002–2011): CFR **8.4%** (PMID: 25605282); Oman 25-year series: **16.7%** (PMID: 42107173); Turkish pediatric cohort: overall **13.3%**, ranging *"from 1.2% in meningitis alone to 21.7% in combined meningitis and meningococcem[ia]"* (PMID: 42802197) (F003). **Shock is the dominant independent predictor of death (aOR 23.30, P<0.0001)** (PMID: 25605282).

Morbidity / sequelae. **~18%** develop complications; the most common in the Canadian cohort were *"hearing loss (5.4%), skin scarring (5.4%), amputation (3.4%), renal dysfunction (2.6%), and seizures (2.5%)"* (PMID: 25605282). A broader US/Canada review found *"10.6-41% of IMD survivors had ≥ 1 sequelae"* — seizures, deafness, stroke, amputation, scarring, renal failure — plus persistent emotional distress in survivors and caregivers (PMID: 40627264) (F004).

Prognostic factors. Presence and depth of shock, DIC, extensive purpura, older age, immunosuppression, and high bacterial/LPS load worsen prognosis; the hemodynamic presentation is associated with older age, immunosuppression, and bacteremia (PMID: 40794167). Prognostic biomarkers: plasma LPS, IL-10, MIF, DNA copy number (PMID: 25233959, PMID: 18679115).

Recovery. With prompt treatment most survivors recover; myocardial depression is typically reversible (PMID: 12394943, PMID: 30871501), but ischemic tissue loss (amputation, scarring) and sensorineural hearing loss are permanent.

Section 12 — Treatment

Empiric/first-line pharmacotherapy. Third-generation cephalosporins — ceftriaxone or cefotaxime — are first-line (NCIT: Ceftriaxone C382; Cefotaxime C328). Surveillance confirms *N. meningitidis* remains uniformly susceptible: *"All isolates were susceptible to ceftriaxone, cefotaxime, meropenem, rifampin, chloramphenicol, and azithromycin. Susceptibility rates were reduced for penicillin (46.8%)"* (PMID: 42107173); US surveillance found 8–16.7% penicillin-intermediate isolates driven by *penA* mutations (PMID: 26357666) (F006). Meropenem (NCIT C1662) and chloramphenicol are alternatives. Meningeal-dose beta-lactam is warranted when CNS involvement is possible even without pleocytosis (PMID: 30871501).

Supportive/critical care. Aggressive **fluid resuscitation, vasopressors** (norepinephrine), respiratory support, and correction of coagulopathy; the majority of severe cases require intensive care (69.9% PICU admission in the Turkish cohort) (PMID: 42802197). For **Waterhouse-Friderichsen syndrome**, **corticosteroid (stress-dose) replacement** plus vasopressors is essential (PMID: 38558654, PMID: 38408808).

Adjuncts under investigation. Vasodilators for DIC-associated distal ischemia showed benefit in a case report and merit study (PMID: 30871501). Anticoagulant/protein-C-pathway–directed therapies are mechanistically rational given EPCR shedding but are not standard.

Chemoprophylaxis of contacts (part of treatment strategy). *"The first-choice antibiotic for chemoprophylaxis of close contacts is ciprofloxacin, followed by rifampicin"* (PMID: 34843604); ceftriaxone is an alternative (NCIT: Ciprofloxacin C405; Rifampin C671). Secondary-case risk is highest in the first days and *"secondary cases rarely occur after 14 days"*, so prophylaxis must be prompt (PMID: 18288075) (F006).

Resistance surveillance. Sporadic ciprofloxacin resistance (*gyrA*) and penicillin/ceftriaxone reduced-susceptibility invasive lineages (serogroup W ST-11) are emerging, mandating ongoing monitoring (PMID: 35467371, PMID: 18288075).

Pharmacogenomics / advanced therapeutics. Not applicable — no gene, cell, RNA, or targeted molecular therapy exists for the infection itself.

Section 13 — Prevention

Primary prevention — vaccination (mainstay). (F008)

Vaccine	Target	Key evidence
MenAfriVac (monovalent A conjugate)	Serogroup A	*"Wide use of the groundbreaking vaccine MenAfriVac® has eliminated serogroup A meningococcal meningitis in the African meningitis belt"* (PMID: 40968077, PMID: 35455366)
Quadrivalent ACWY conjugate (MenACWY-TT, MenACWY-CRM)	A, C, W, Y	Adolescent programs; infant safety demonstrated (PMID: 42642862)
Pentavalent ACWYX (NmCV-5/MenFive)	A, C, W, Y, X	*"Crude VE against all meningococcal meningitis was 88% (95% CI: 62-96), and against NmC specifically was 86% (95% CI: 50-96)"* in a Niger outbreak (PMID: 42585809, PMID: 40968077)
4CMenB (protein-based)	Serogroup B	England: 75% reduction in MenB IMD in immunized infants; adjusted efficacy 52.7% after primary series, 59.1% after booster (PMID: 42347623)

Serogroup replacement (e.g., emergence of X, shifts in C/W) drives the move to broader pentavalent coverage — *"Serogroup X showed a significant upward trend ($P < 0.01$)"* (PMID: 37024833).

Behavioral primary prevention. Smoking cessation and reduction of household crowding lower carriage/transmission (PMID: 11399012).

Secondary/tertiary prevention. Rapid recognition of red-flag symptoms and immediate antibiotics prevent progression to shock; **contact chemoprophylaxis** (ciprofloxacin/rifampicin) prevents secondary cases (PMID: 34843604, PMID: 27227736).

Special high-risk population — complement-inhibitor recipients. Anti-C5 therapy (eculizumab, ravulizumab) *"increases the risk of invasive meningococcal infections"* by blocking terminal C5b-9 (PMID: 42137634, PMID: 42234874) (F009). Expert consensus mandates **vaccination against A, C, W, Y and B for all patients on complement inhibitors, plus antibiotic prophylaxis** (penicillin/rifampicin), because **breakthrough IMD occurs despite vaccination** (PMID: 42074848).

Public-health interventions. Outbreak response includes mass vaccination and expanded chemoprophylaxis of contacts (PMID: 27227736); surveillance of serogroups and resistance guides policy.

Section 14 — Other Species / Natural Disease

***Neisseria meningitidis* is a strictly human-specific pathogen** with **no natural animal reservoir or naturally occurring animal disease** (F013). *"Studying the pathogenesis of this infection was previously limited by the human specificity of the bacteria, which makes in vivo models difficult"* (PMID: 24747976). Consequently:

- **Taxonomy of affected species:** *Homo sapiens* only (NCBI Taxon 9606). Pathogen: *Neisseria meningitidis* (txid487).
- **Breed / VBO:** not applicable.
- **Natural disease in other species (OMIA/VetCompass):** none — there is no companion-animal or wildlife meningococcemia.
- **Comparative biology:** the human specificity arises from species-restricted host receptors (e.g., human EPCR, human complement regulators, human CEACAMs and transferrin) that the meningococcus exploits. This is a striking example of host-restricted molecular tropism.
- **Zoonotic potential:** none.

Section 15 — Model Organisms

Because of strict human specificity, conventional animal models fail; the field relies on **humanized systems** (F013):

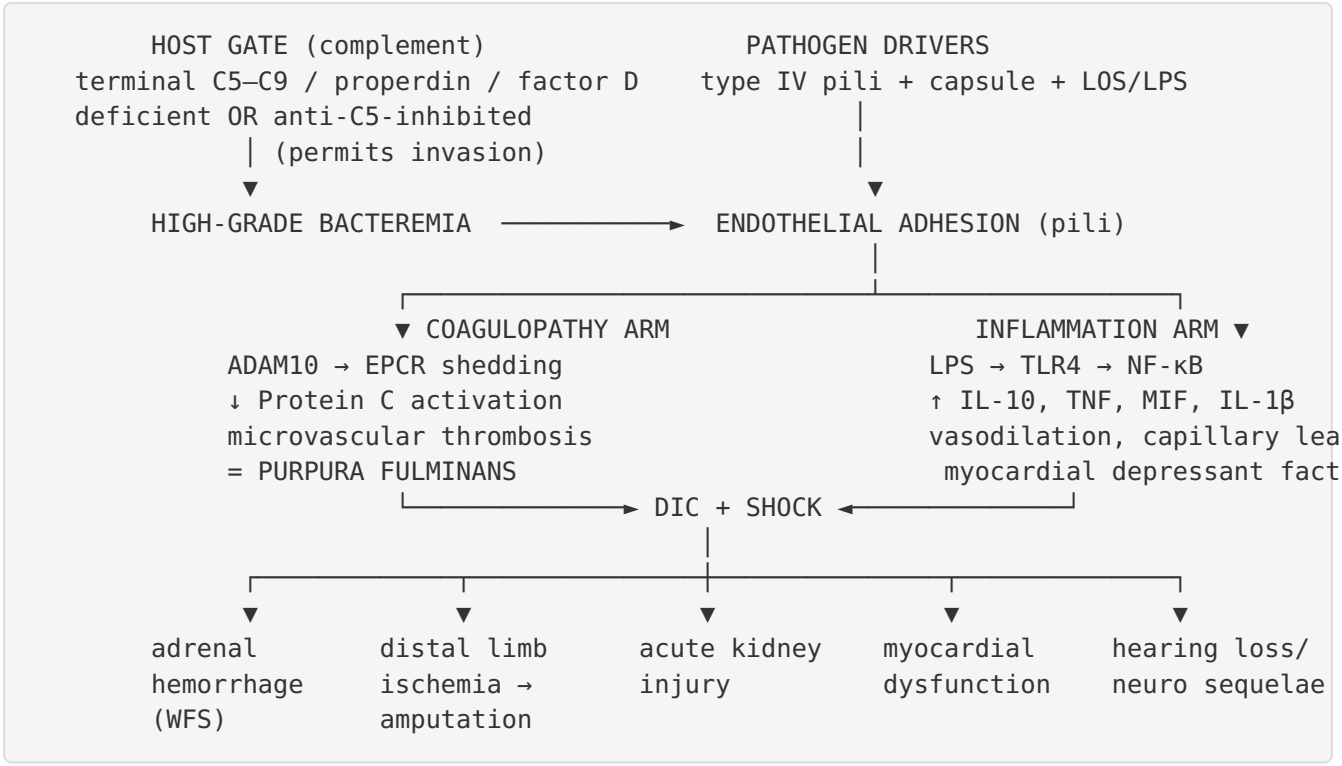
Model	Type	Recapitulation	Reference
Human skin-graft SCID/Beige mouse ("humanized mouse")	Mammalian, humanized xenograft	Bacteria adhere exclusively to human dermal microvessels → human cytokine secretion, inflammatory recruitment, vascular leakage, thrombosis, and **purpura** — *"replicating the clinical presentation for the first time in an animal model"*	PMID: 23359320, PMID: 24747976
Type IV pilus mutants in the humanized model	Genetic (bacterial) manipulation	Pilus-adhesion mutants fail to cause vascular damage; **PilT-dependent pilus retraction required for sustained bacteremia and lethality**	PMID: 33592056
3D microfluidic human vessel model	In vitro engineered human vasculature	Reproduces endothelial integrity/ permeability; quantifies microcolony growth, cytoskeleton rearrangement, E-selectin expression, neutrophil response; validated against skin-xenograft mouse	PMID: 41406015
Properdin-knockout mouse	Mammalian genetic	Probes complement/shock biology in non-septic shock models (not a full IMD model)	PMID: 18453618

Key mechanistic insight from models: type IV pili adhesion is the main in vivo mediator of vascular association (*"The adhesive properties of the type IV pili of N. meningitidis were found to be the main mediator of association with the dermal microvessels in vivo"*, PMID: 23359320).

Limitations of models: the skin-graft model captures the microvascular/purpuric arm but not full systemic multi-organ sepsis, meningitis, or the human adaptive immune response; in vitro vessel models lack systemic physiology. No standard MGI/ZFIN/FlyBase genetic model of the disease exists because the host substrate must be human tissue.

Mechanistic Model / Interpretation

The unifying interpretation is that **meningococcemia is a disease of two converging catastrophes on the human microvascular endothelium**, gated upstream by complement.



Two features make meningococcemia mechanistically distinctive rather than generic Gram-negative sepsis: (1) the **specific ADAM10/EPCR/Protein-C lesion** that concentrates thrombosis in the skin microvasculature to produce purpura fulminans, and (2) the **complement gate** that determines who gets invasive disease at all — a gate now routinely opened pharmacologically by anti-C5 drugs. Upstream = complement gate + bacteremia; midstream = pili-driven endothelial adhesion; downstream branches = coagulopathy (EPCR) and inflammation (LPS) converging on DIC/shock; terminal = organ-specific damage and, in survivors, permanent ischemic sequelae.

Evidence Base

Finding	Key PMIDs	Evidence type	Role
Complement/properdin/ factor D deficiency → susceptibility	25037634, 16527897, 16337490, 10909851	Human genetics/ immunology	Supports host gate
EPCR shedding / purpura fulminans mechanism	29630665, 33592056	In vitro + humanized model	Supports coagulopathy arm
Two clinical forms, form-dependent mortality	40794167, 42802197	Human cohorts	Clinical phenotype
Mortality ~8–17%, sequelae 10–20%	25605282, 40627264, 42107173	Human cohorts/ reviews	Prognosis
LPS/IL-10/MIF drive shock	25233959, 18679115, 17438452, 12394943	Human + in vitro	Inflammation arm
Cephalosporin first-line; prophylaxis; resistance	42107173, 26357666, 34843604, 18288075, 35467371	Surveillance/ guidelines	Treatment
Carriage/transmission risk factors	11399012, 27861618, 11057965	Human epidemiology	Etiology/ prevention
Conjugate & 4CMenB vaccines	40968077, 42585809, 42347623, 35455366, 37024833	Trials/ effectiveness	Prevention
Iatrogenic risk from anti-C5	42234874, 42137634, 42074848	Consensus/case reports	Etiology/ prevention
Waterhouse-Friderichsen syndrome	42665693, 38408808, 38558654	Autopsy review/ cases	Complication
Diagnosis: culture + ctrA PCR	11283086, 10799809	Diagnostic validation	Diagnostics
Early red-flag symptoms	21375891	Primary-care study	Phenotype/ diagnosis
Human-restricted; humanized models	23359320, 24747976, 33592056, 41406015	Model organism	Models/species
			Epidemiology

Finding	Key PMIDs	Evidence type	Role
Epidemiology: age, season, disparity	16511378, 42802197, 18322129	Population surveillance	

Limitations and Knowledge Gaps

1. **No primary molecular data were analyzed** in this investigation; findings are a literature synthesis. Quantitative claims derive from the cited cohorts and reviews, whose populations, eras, and definitions vary (pediatric vs adult, endemic vs epidemic settings), so pooled mortality/sequelae ranges are approximate.
2. **Serogroup- and region-specific granularity** (variant-level epidemiology, exact incidence per serogroup by country/year) was not exhaustively resolved; serogroup replacement is dynamic.
3. **Host-genetic susceptibility beyond complement** (e.g., GWAS-identified common variants such as CFH/CFHR3 loci) was not deeply explored and merits dedicated review.
4. **Therapeutic modulation of the EPCR/Protein-C axis** is mechanistically compelling but clinically unproven; whether restoring EPCR/activated Protein C alters purpura fulminans outcomes is unknown.
5. **Model limitations:** humanized skin-graft and microfluidic models capture microvascular pathology but not systemic multi-organ sepsis, meningitis, or adaptive immunity.
6. **Long-term neurocognitive and psychosocial outcomes** in survivors are under-quantified relative to acute mortality.

Proposed Follow-up Experiments / Actions

1. **Test EPCR/Protein-C–directed adjunct therapy** (e.g., recombinant activated Protein C analogs, ADAM10 inhibition, soluble EPCR) in the humanized skin-graft and 3D microfluidic vessel models, then in early-phase trials for purpura fulminans.
2. **Prospective biomarker-guided risk stratification**: validate a combined plasma LPS + IL-10 + meningococcal DNA copy-number panel as an early triage tool to identify patients destined for shock (PMID: 18679115).

3. **Systematic host-genetic study** integrating complement functional assays (CH50/AH50) with sequencing of terminal-complement, CFP, CFD, MBL2, and complement-regulator genes in recurrent/atypical IMD survivors to define screening yield.
4. **Vaccine-policy modeling** for a life-course strategy (infant + adolescent + older-adult) and for pentavalent ACWYX rollout, incorporating serogroup-replacement surveillance (PMID: 42702664, PMID: 37024833).
5. **Optimize complement-inhibitor safety**: prospective registry of breakthrough IMD in eculizumab/ravulizumab recipients to define the optimal vaccination + antibiotic-prophylaxis regimen (PMID: 42074848).
6. **Antimicrobial-resistance genomic surveillance** for *penA*/*gyrA* lineages (serogroup W ST-11) to preserve empiric cephalosporin and prophylaxis efficacy (PMID: 35467371).
7. **Trial of vasodilator therapy** for DIC-associated distal ischemia to reduce amputation, following the case-report signal (PMID: 30871501).

Report compiled from 14 confirmed findings and 57 reviewed papers across a 5-iteration investigation. Ontology suggestions (HPO, GO, CL, UBERON, NCIT, MONDO) are provided inline for knowledge-base ingestion.
